

Instructor: ..... Date: .....

Learner: ..... Regional Branch: .....

Program: ..... Center: .....

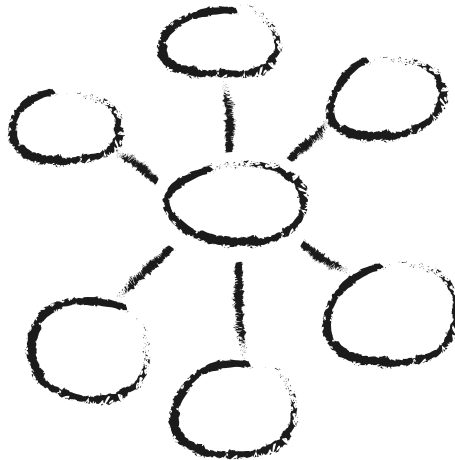
## CONTENT

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### Let's Start!

It's time to design data warehouse!



### Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.  
Understand information about data and its structure in simple spoken and written situations.

### Can Do Statements:

- I can talk about data warehouse design.
- I can explain key features related to data warehouse design.
- I can give specific information related to data warehouse design.

### Ask your partner:

- What do you consider important to design a database?
- In your opinion, what are the basic elements of a data warehouse?





### Let's Explore!

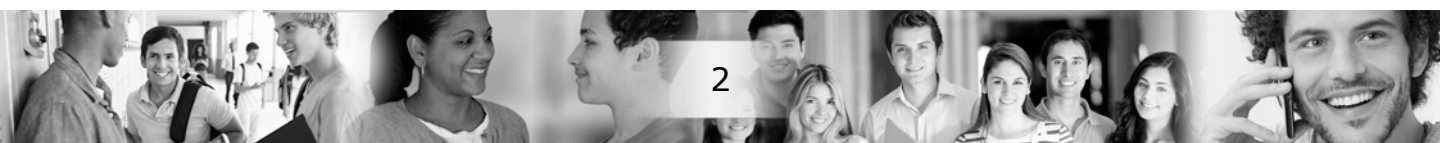
David, Neal, and Roxana are defining the schema they are going to use to build their data warehouse.

\*"Take into account" is an expression that means to consider.



*Instructions: Read the text below.*

- David:** Ok, guys. We need to finish the schema of how our data warehouse is going to look and behave if we don't want to have future problems with it.
- Neal:** To start with, we can set it as read-only so that no one can modify the data. Anyway, what kind of design are we going to work with?
- David:** That's a very good question, Neal. And that's why I called for this meeting because if we don't define a schema today, the possibilities to move forward with the project are almost null. Thus, here I have 2 possible designs we can work with if you don't mind. The first one is the star schema, and the other option is the snowflake one.
- Roxana:** Sorry to interrupt, but even though we have the schema, I think the very first thing we need to define is the factors that take part in it; for example, facts, measures, dimensions, and hierarchies. If we don't take them into account, the schema is useless.
- David:** Roxana, I see some new faces here, why don't you explain to them what those factors are about.
- Roxana:** Sure! Guys, remember that a fact in a data warehouse scheme is the focus of it; you know, things like sales or purchases. Additionally, the measures are the same attributes for each fact, for example, the revenues or profits from the sales. Similarly, we talk about dimensions, which are the minimal parts in which we can divide a fact, such as dates, products or stores. On the other hand, we have the hierarchies, which are the same dependencies we have in the company, like the sales or the HR departments, which, by the way, don't share their data if they don't have a data mart in their dependencies.
- David:** Thanks, Roxana. In order to continue with our agenda...





## Let's Practice!

Follow these steps to design a data warehouse

*Instructions: Match the sentences from Column A with Column B. Read the sentences aloud. Check your pronunciation with your instructor.*

### COLUMN A

1. Even though we have the schema, I...
2. There are some factors that take part in the data warehouse design;
3. A fact in a data warehouse scheme is the focus of it.
4. Similarly, we talk about dimensions, which...

### COLUMN B

- a. are the minimal parts in which we can divide a fact, such as dates, products or stores.
- b. for example, facts, measures, dimensions, and hierarchies.
- c. Additionally, the measures are the same attributes for each fact.
- d. think the very first thing we need to define is the factors that take part in it.



## Let's Practice!

What do you have to take into account when designing a data warehouse?

*Instructions: Select the correct answer in order to complete the sentence.*

1. The sales and HR departments don't usually share their data if they don't \_\_\_\_\_ a data mart in their dependencies.  
(hint: a verb expressing possession)  
a. have  
b. possess  
c. take
2. The legacy databases do not \_\_\_\_\_ with the data warehouse If you don't layout how they are supposed to communicate.  
(hint: Synonym to connect)  
a. communicate  
b. talk  
c. contacts
3. If you don't \_\_\_\_\_ a snowflake schema, use the star design.  
(hint: to choose something)  
a. user  
b. make use  
c. opt to use
4. It is an ideal data warehouse \_\_\_\_\_ let you modify the data.  
(hint: take care with the subject.)  
a. if he doesn't  
b. if they don't  
c. if it doesn't
5. If \_\_\_\_\_ what schema to use, look at the need of your company.  
(Hint: to be aware of something)  
a. you don't know  
b. you don't care  
c. you don't mind





### Extension Activity

Now's your turn to design your own data warehouse.

*Instructions: Complete the following steps. Fill in the Self-Assessment.*

- Read more about data warehousing design.
- Describe the steps to follow to design one for a mid-size company.
- Write a short text (100 words) explaining briefly each of the designs and their importance, remember to use the grammar of the lesson.
- Discuss your work with your partners.

#### Self-Assessment

- I get information about data warehousing design.
- I list the steps to follow to design it.
- I write a short text explaining briefly each one of the designs and their importance, using the grammar of the lesson.
- I discuss my work with my partners.

Yes No

☐ ☐
☐ ☐
☐ ☐
☐ ☐


### Target Words

- revenue** (noun): the income of a company  
e.g. Additionally, the measures are the same attributes for each fact, for example, the revenues or profits from the sales.
- thus** (connector): for that reason; therefore  
e.g. The possibilities to advance with the project are almost null. Thus, here I have 2 possible designs we can work with if you don't mind.
- call for something** (expression): to eagerly request for something  
e.g. That's a very good question Neal, and that's why I called for this meeting because if we don't define a schema today, the possibilities to advance with the project are almost null.
- schema** (noun): a layout or a design of something  
e.g. We need to finish the schema of how our data warehouse is going to look like if we don't want to have future problems with it.
- null** (adjective): useless; worthless  
e.g. The possibilities to advance with the project are almost null.
- HR department** (acronym): Human Resources Department  
e.g. The hierarchies, which are the same dependencies we have in the company, like the sales or the HR department.



## Learning Points

### Connectors review

Use **connectors** to link words, phrases and sentences.

Remember that connectors are words you use to link words, phrases and sentences in a text. Remember that connectors can be **coordinating conjunctions**, such as **and**, **or**, **and but**, that connect items that belong to a same grammatical type.

Examples:

*We need to implement a data warehouse **and** a data mart for each dependency of the company (noun and noun)*

*We don't know if the database is useless **or** outdated. (adjective and adjective)*

*We don't need to change the database **but** to update it. (verb and verb)*

Remember that conjunctions do not always need a comma before them, but connectors such as linking adverbs need a comma, a semicolon or a period before them.

Linking adverbs show a mental relationship between two sentences or clauses. Let's look at some of them.

**Contrast:** by using these kind of connectors, you emphasize differences between ideas; some of these connectors are the following: **on the other hand**, **by comparison**, **however**, **in contrast**.

Example:

*Samantha knows how to work with data warehouses. **However**, she is out of the country right now.*

**Similarity:** You use these types of connectors when you compare and add information, for example, **in the same way**, **also** and **similarly**.

Example:

*Data warehouses keep all the information of the company in one place. **Similarly**, data marts compile the data of each dependency of the company.*

**Result:** You can use these connectors to show the consequences or the conclusions of a topic. Among these kind of connectors, we find **as a result**, **therefore**, **as a consequence** and **accordingly**.

*Data warehouses compile all the company's data; **therefore**, you need to allow all data marts to share data with the warehouse.*





## Learning Points

**Sequencing:** You can use them to give an order and coherence to your ideas, for example, **to continue, to begin with, and finally.**

Example:

**To begin with,** *there are many ways to keep your data sound and safe without the use of data warehouses...*

### Examples

- *To start with, I agree with you. However, what kind of design are we going to work with?*
- *Thus, here I have 2 possible designs we can work with if you don't mind.*
- *Even though we have the schema, I think the very first thing we need to define is what factors are taking part in it.*

## Zero Conditional with the if clause in negative

Use **zero conditional** to express information about specific situations that commonly have a cause and effect.

In order to analyze the zero conditional, we must know what the conditional mood is.

We can see the conditional mood as a set of sentences showing the relation between facts (real or hypothetical) and their conditions.

Therefore, a conditional sentence must have two clauses or ideas, a main clause and a conditional clause (if clause).

We can use the zero conditional with the if clause in negative form to express ideas that even though are always true, have a negative condition.

Remember that both ideas (clauses) in zero conditional have the verbs in present form; the conditional clause goes with "if" or "when," and the main clause goes in simple present.

Structure:

1. If/when + present simple negative + , + present simple,
2. Present simple + if/when + present simple negative.





## Learning Points

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Examples:

- if you don't want to share the data stored in the data mats with the data warehouse, you need to carry more time consuming processes to keep track of it.
- Store all the data of the company in the data warehouse if you don't want to deal with unrelated data problems in a future.

Notice that when the if clause goes first in order, you need to add a comma “,” to separate both ideas; but when you start the idea with the main clause, you don't need to add a comma to separate the ideas.

### Examples

- *Here I have 2 possible designs we can work with if you don't mind.*
- *If we don't take them into account, the schema is useless.*
- *If you don't consider something else is missing thus far, we need to choose the design and its components right away.*